HARJOT SINGH

AI/ML Engineer · Computer Vision Specialist · Edge AI

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OBJECTIVE

Second-year B.Tech AIML student at Thapar Institute of Engineering & Technology with proven, hands-on experience building production-grade Computer Vision systems, real-time object detection pipelines, and large-scale data engineering solutions. Recognised as a Top 17 Finalist at SATHACK'25. Seeking an internship where I can apply deep learning and edge AI expertise to ship impactful products alongside a forward-thinking engineering team.

EDUCATION

B.Tech — Artificial Intelligence & Machine Learning · *Thapar Institute of Engineering & Technology* 2024 – Present
Patiala, India · Currently pursuing

Class XII — CBSE · *Modern Senior Secondary School* 2024
Patiala, India · 83.4%

Class X — ICSE · *Kaintal School* 2022
Patiala, India · 91.4%

TECHNICAL SKILLS

Languages	Python · C++ · JavaScript
AI / ML	Machine Learning · Deep Learning · CNN · Transfer Learning · Transformers
Computer Vision	YOLO · OpenCV · Object Detection · Image Processing
Edge AI	Real-time inference · Model optimisation · Embedded deployment
Data & Tools	Pandas · NumPy · Docker · Git & GitHub · Jupyter Notebook
Web Dev	HTML · CSS · JavaScript

PROJECTS

Smart Traffic Monitoring System Python · OpenCV · YOLO · Transformers

- Engineered a real-time vehicle detection and multi-class classification pipeline using YOLO on live camera feeds, achieving sub-second inference latency.
- Built a traffic density analysis engine that identifies and quantifies congestion patterns to generate actionable, time-stamped insights.

- Designed an end-to-end pipeline from raw video ingestion through inference to structured actionable traffic insights.

X-Ray Dangerous Object Detection **Python · YOLOv8 · Computer Vision**

- Developed a deep learning pipeline for automated, high-throughput security baggage scanning — directly reducing manual inspection workload.
- Fine-tuned YOLOv8 for multi-class threat detection, optimising precision–recall trade-offs for security-critical environments.
- Achieved high-accuracy real-time inference suitable for deployment at airport and venue security checkpoints.

Traffic Challan Data Engineering Pipeline **Python · Pandas · Docker**

- Built a scalable ETL pipeline that ingests, cleans, and transforms 416M+ Indian traffic e-challan records — one of the largest open civic datasets in India.
- Implemented robust data cleaning and aggregation stages, producing downstream-ready datasets and analytical dashboards surfacing violation trends by region and vehicle type.
- Containerised the entire pipeline with Docker, ensuring full reproducibility and one-command execution across environments.

ACHIEVEMENTS & EVENTS

Top 17 Finalist — SATHACK'25 Hackathon — Nov 2025

Thapar Institute of Engineering & Technology · Saturnalia 50th Anniversary

ELC Summer Internship — Jun – Jul 2025

Experiential Learning Centre -Thapar Institute of Engineering & Technology

CURRENT FOCUS

Advanced Computer Vision Systems · Real-time AI Applications · Transformer Models · Scalable ML Pipelines · Edge AI Deployment